Read me first

BlackSwanQuants: Take-home project (returning contributors track)

Welcome, and thanks for doing this.

What this is, and what it is not

You are going to spend a single day building a small, honest piece of quant research on a **pair of related assets** (your choice of pair, see the task below). The task is deliberately larger than what anyone can finish in the time given. That is on purpose.

We are not testing whether you can build a profitable strategy. A backtest that makes money on one specific slice of history is easy to produce and worth very little on its own. **What we are testing is whether you can tell a real signal from a lucky accident**: whether your validation is honest, whether you know how many things you tried before landing on your result, and whether you can defend your choices out loud afterwards.

If you finish early and are unsure what to do with the extra time, look at the "Optional extensions" phases at the end of this document rather than polishing what you already have.

How to approach the day

- **Every phase that lists more than one method is a menu, not a checklist.** Pick **one** method per phase: the one you can execute well and defend, not all of them half-done. A single method done rigorously beats three done superficially.
- **Do a few things well rather than many things half-way.** We would rather see a solid, honestly validated signal on half the phases than a sprawling notebook that falls apart on a second look.
- **Say when you are unsure.** "I only tried 3 parameter combinations, here they are" is a strong answer. Hiding how many things you tried, or how you picked the one you show us, is the one thing that actually costs you points.

The practicalities

Work window	**12 hours**, self-paced, within a single day. Take breaks.
Data	Both the price data and the supplementary data in Phase 4 are freely available; see the task below for exact sources. Ask us if you get stuck sourcing something: logistics questions are always fine.
Stack	Completely your choice. Python is the natural fit given the ecosystem (pandas, numpy, statsmodels, a plotting library), but nothing is mandated.
AI coding assistant	**Strongly suggested.** Driving an AI agent well is a real skill and part of what we are evaluating: treat it as a core part of the exercise. See the disclosure rule below.

What to deliver

1. **A notebook or a small tidy folder of scripts**, organized so a stranger can follow it phase by phase.
2. **A short writeup** (half a page is enough): which phases you completed, which method you picked where there was a choice and why, what you would do with more time.
3. **AI usage disclosure (always required, not conditional):** leave in the folder any supporting files your AI assistant produced (scripts, notes, generated data), as-is. And regardless of whether it produced standalone files, always write a short paragraph on how you used it: what you asked it to do, what you checked yourself, and where you overrode or corrected it. How well you drove it is itself part of what we are evaluating.

That disclosure is not a formality: it is one of the clearest windows we have into how you actually work.

How we evaluate

In order of weight: **how honestly you validated your findings** (this is the main thing), **the soundness of your reasoning** at each step, **the quality of how you present your results** (a clear plot beats a wall of numbers), and, last, whether the strategy or pattern you found actually made money. A candidate who found nothing but can explain exactly why, and how hard they tried to find something real before giving up, is in great shape.

Good luck, we are looking forward to seeing what you build.

Take-home project

Two assets, one relationship, and how hard you tried not to fool yourself

Work window: 12 hours, self-paced, within a single day.

Assets: pick **one pair** and stick with it for the whole project. Three suggestions, all with 50+ years of free daily history on Yahoo Finance, so data depth is never the bottleneck:

- **KO / PEP** (Coca-Cola / PepsiCo): the classic textbook pair, same sector.
- **XOM / CVX** (ExxonMobil / Chevron): two oil majors, same sector.
- **INTC / AMD** (Intel / AMD): the classic x86 CPU rivalry, same sector. Both are Nasdaq-listed, so this is the one pair of the three where the short interest source in Phase 4 works cleanly on both legs (see below).

You may pick a different pair if you prefer, as long as the two assets are plausibly related (same sector, same commodity exposure, etc.) and you can get at least ∼15–20 years of free daily data for both from Yahoo Finance (ticker in hand, use `yfinance` or a manual download).

0. The spirit of this exercise

Every phase below that offers more than one method is a **menu**: pick **one** option, the one you can do well and explain, not all of them at once. The scope is bigger than what fits in 12 hours on purpose: we want to see what you choose to build and what you choose to skip.

The single most important thing we are looking at is **Phase 2**: not whether your signal makes money, but whether your validation of it is honest. Everything else in this brief exists to give you material to reason about; Phase 2's standard applies to every phase that involves testing something.

Unlike a strategy that bets on the direction of one asset, here you are trading the **relationship between two assets**: you make money when that relationship reverts to normal after a temporary divergence, regardless of which way the overall market moves. That is the core idea behind everything that follows.

1. Data (warm-up)

Download the daily price history for your chosen pair from **Yahoo Finance** (`yfinance` or manual CSV). Plot both price series together, plot the ratio or difference between them over time, and look at the correlation of their daily returns. Get a feel for whether the relationship looks stable or whether it visibly drifts over the decades.

2. A mean-reversion signal on the spread, and its validation

Building the spread

The two prices individually behave like a random walk: they wander, with no fixed level to return to. But if the two assets are genuinely related, a specific *combination* of them can behave very differently: it can hover around a stable average and pull back whenever it strays too far, instead of wandering off forever. That combination is called the **spread**, and finding it is the first real step of this project.

The raw difference $\text{Price}_A - \text{Price}_B$ usually will not do the job on its own, because the two assets do not move one-for-one: one of them might typically move twice as much as the other for a given market move. To fix that, run an **OLS linear regression** of one price on the other:

$$\textit{Price}_A = \alpha + \beta \times \textit{Price}_B + \textit{residual}$$

The estimated β **is called the hedge ratio**: it says "for every €1 move in B, A tends to move €β". Once you have it, define:

$$\textit{Spread} = \textit{Price}_A - \beta \times \textit{Price}_B$$

If β is well chosen, this spread should oscillate around a roughly stable mean instead of drifting like the raw prices do. This is the property you are hoping for, but running the regression does not *guarantee* it; you should actually check it (a basic way: eyeball the spread plot, or look at whether it keeps crossing back over its own historical average repeatedly rather than trending away from it).

The signal

Standardize the spread into a **z-score** (how many standard deviations away from its own rolling average it currently is). Go long the spread (long A, short β units of B) when the z-score is very negative, short it when very positive, and close the position when it reverts back toward zero.

Validation: the part we weight most heavily

Same standard as any strategy:

- **Split your data into an in-sample period (where you look and tune) and an out-of-sample period (where you do not look until you are done).** Report performance on both, separately.
- **Be explicit about how many things you tried.** If you tested several different z-score entry thresholds before picking one, say so ("I tried entry at 1.5σ, 2σ, 2.5σ"): that count matters more than which one you show us. With enough attempts, something will look good on any fixed slice of history purely by chance, whether or not there is a real effect underneath.
- **Check how sensitive your result is to small changes in the parameters** (threshold, lookback window for the rolling mean). If a tiny change collapses the result, that is a sign you found noise, not a real effect; say so.
- **Do not optimize on the out-of-sample period.** If you peek at it and then go back and change something, it stops being out-of-sample.

We would much rather read "I tried 5 things, none of them survived the out-of-sample check, here is my honest read on why" than an unquestioned equity curve that goes up and to the right.

3. Is the relationship stable?

The hedge ratio you estimated is only an estimate: the true relationship between the two assets can drift or break down entirely over time (a merger, a change in one company's business, a structural shift in a sector). Pick **one** way to check for this:

- **Simple:** compute a rolling correlation between the two assets' daily returns (e.g. over a 60–120 day window). Flag periods where it drops well below its historical norm as "weak relationship" periods.

- **Medium:** re-estimate the hedge ratio itself on a rolling window (instead of once on the whole history) and/or re-run a **stationarity check on the rolling spread** (a standard one is the Augmented Dickey-Fuller test; most stats libraries have it built in, and it tells you, with a p-value, how confident you can be that a series is mean-reverting rather than drifting). Flag windows where the spread stops looking stationary.

Then check: does your Phase 2 signal perform worse during the periods you flagged as "weak relationship"?

4. Institutional positioning: short interest

Short interest is the total number of shares of a stock currently sold short (borrowed and sold, betting the price falls), settled and published twice a month. A rising short interest tells you speculators are increasingly betting against a stock; a falling one, the opposite.

A real constraint to navigate, not a trick: the cleanest free, no-login source is Nasdaq's own public data (`api.nasdaq.com/api/quote/<TICKER>/short-interest`, plain JSON, no key needed), but it **only covers Nasdaq-listed tickers**, not NYSE-listed ones. Check which of your two assets actually qualifies before you rely on it:

- If you picked **KO / PEP**: only **PEP** is Nasdaq-listed; KO (NYSE) will not have data through this source. You can still do the analysis on PEP's short interest alone against the pair's spread/regime.
- If you picked **XOM / CVX**: **both are NYSE-listed**, so this particular free source has nothing for either leg. If that is your pair, say so plainly, and either look for an alternative source yourself or explain why you are skipping this phase: both are legitimate, gradeable outcomes.
- If you picked **INTC / AMD**: **both are Nasdaq-listed**, so you can pull short interest for both legs through this source with no workaround needed.
- If you picked your own pair, check both tickers against this constraint first.

Depth is also limited on purpose by the source itself (a rolling $\sim$12 months, updated twice a month), much shorter than your price history. Work with whatever window and coverage you actually get.

For whichever ticker(s) and window you can get: does a change in short interest line up with anything from Phases 2–3 (a divergence in the spread, a period the relationship weakens, a bigger-than-usual move)? A clean "nothing showed up in the data I could get, here is why I believe that", or even "this data was not available for my pair at all, here is what I would have looked for if it were", is a fully valid finding.

5. Simulating the pair

Real market history is a single path: only one thing actually happened to this pair. Simulation lets you generate other plausible joint paths so you can ask "would my Phase 2 signal still have looked good if history had gone slightly differently?" You need to simulate **both series together**, preserving how they move together, not simulate each one separately. Pick **one**:

- **Simple: joint block bootstrap.** Take the historical daily returns of **both** assets and resample **the same time blocks, for both series at once** (e.g. blocks of 5–20 consecutive days), with replacement, stitching the chunks together into a new synthetic joint path. Taking the same block from both series together (not separately) is what preserves the correla-

tion between them, along with each series' own short-term patterns like volatility clustering. Generate several synthetic pairs of paths and re-run your Phase 2 signal on each: how consistent is its performance across them?

- **Medium: simulate the spread directly with a mean-reverting process.** Since you already believe (from Phase 2) that the spread tends to revert to a mean, you can model it directly as an **Ornstein-Uhlenbeck process**, a standard way to describe something that drifts randomly but is continuously pulled back toward a long-run average, at a speed and with a volatility you estimate from your historical spread. Simulate several synthetic spread paths from that calibrated process (no need to simulate the two raw prices separately) and re-test your z-score signal on them.

Report how your signal's performance varies across the simulated paths. This is your best evidence for or against the relationship being a real, robust effect rather than a fit to one specific historical accident.

6. Optional extensions (only after the core is solid)

Pick freely, in any order, only if you have time left after Phases 1–5:

- **Dynamic hedge ratio with a Kalman filter:** instead of a single static OLS regression in Phase 2, use a Kalman filter to let the hedge ratio update continuously over time, and see if it improves on the static version.
- **A basket instead of a pair:** extend to 3+ related assets instead of 2, and build a spread from all of them together.
- **Transaction costs and walk-forward:** add a realistic per-trade cost to your Phase 2 backtest, and replace the single in-sample/out-of-sample split with a rolling walk-forward validation.
- **A second pair:** repeat the analysis on a second, different pair and see if what you found generalizes.

7. What to deliver

1. A notebook or a small tidy folder of scripts, organized phase by phase.
2. A short writeup (half a page): which phases you completed, which method you picked wherever there was a choice and why, what you would do with more time.
3. AI usage disclosure, as described at the top of this document.

Good luck.